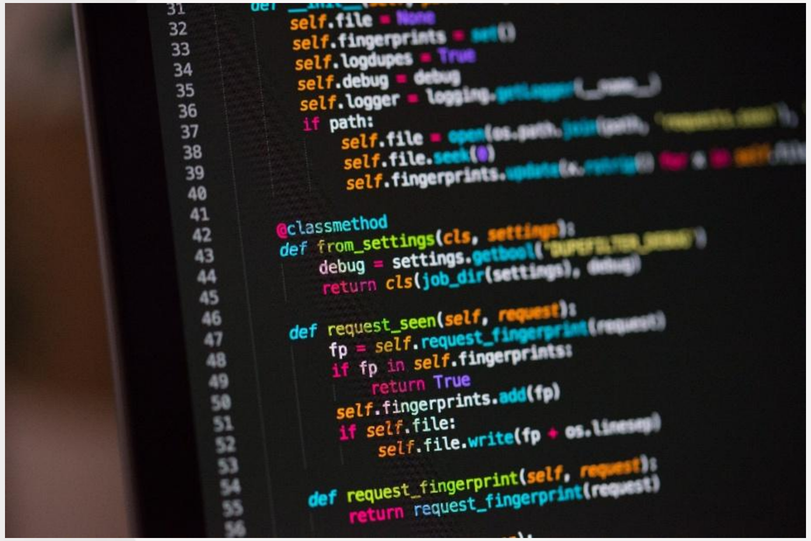
# Level 5 Data Engineer
# Module 6 Topic 2

## Heterogeneous Data Ingestion Patterns




**L5 Data Engineer Higher Apprenticeship**
Module 6 / 12 (**“Data Collection and Ingestion pt. 1”**)
Topic 2 / 4

# Ice breaker: Discussion

A bit of fun to start…

1. If you could automate any mundane task in your daily life using an ETL process, what would it be and why?

2. Imagine you could have a metadata catalog for your personal life. What kind of information would you tag and why?

3. How do you think effective metadata management can improve data quality and compliance in an organisation?

**Submit your responses to the chat or turn on your microphone**

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# Case study

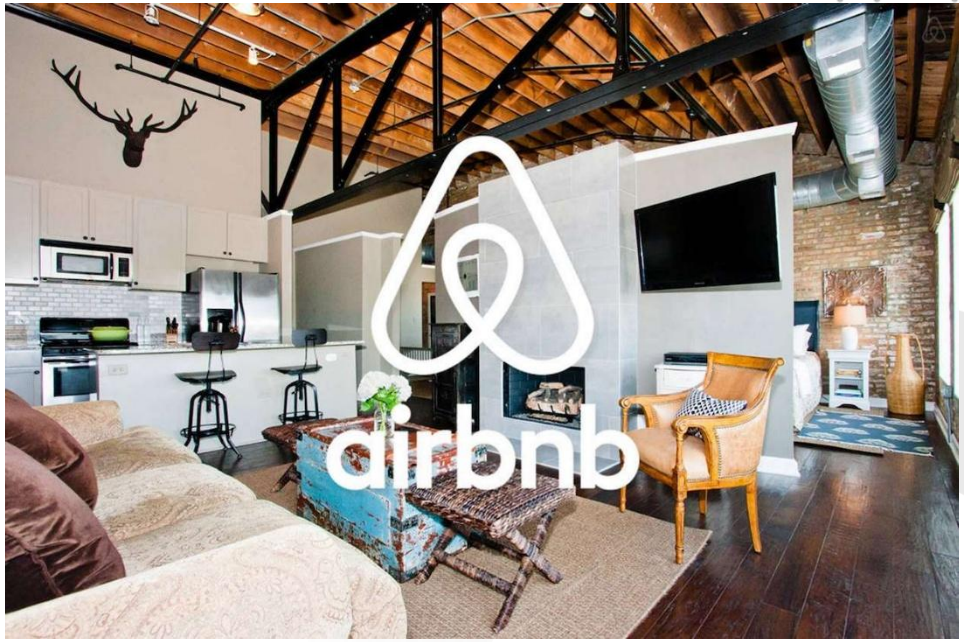
## Airbnb’s Use of ETL and Metadata Management

Airbnb uses ETL processes to handle massive amounts of data from various sources, including:

- User interactions
- Booking details
- Property listings
- Employs metadata management to ensure data quality and compliance

**Impact:**

- Provides personalised user experiences
- Optimises search results
- Maintains high data quality standards
- Crucial for business operations

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# Knowledge check poll

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What is the primary benefit of using ETL (Extract, Transform, Load) processes in data ingestion?

A. It allows for real-time data processing.

B. It reduces the need for data storage.

C. It eliminates the need for data validation.

D. It ensures data is cleaned and transformed before loading into the data warehouse.

**Submit your responses to the chat!**

**Feedback: D –** It ensures data is cleaned and transformed before loading into the data warehouse.

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# Session aim and objectives

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Completion of this topic supports the following outcomes:

- Explain the key design and architecture considerations of heterogeneous data ingestion systems
- Report on the usefulness of heterogeneous ingestion patterns for business use cases
- Demonstrate practical application of file ingestion combining multiple sources of data

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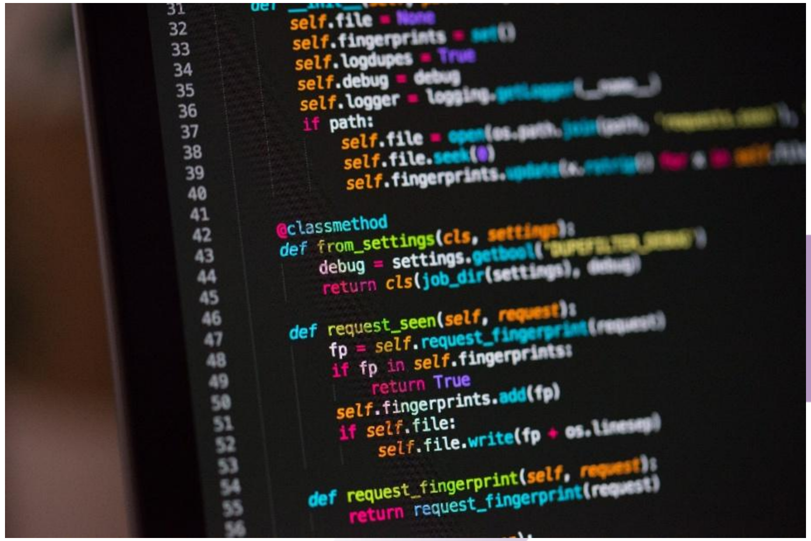
# E-learning Recap



# Recap discussion

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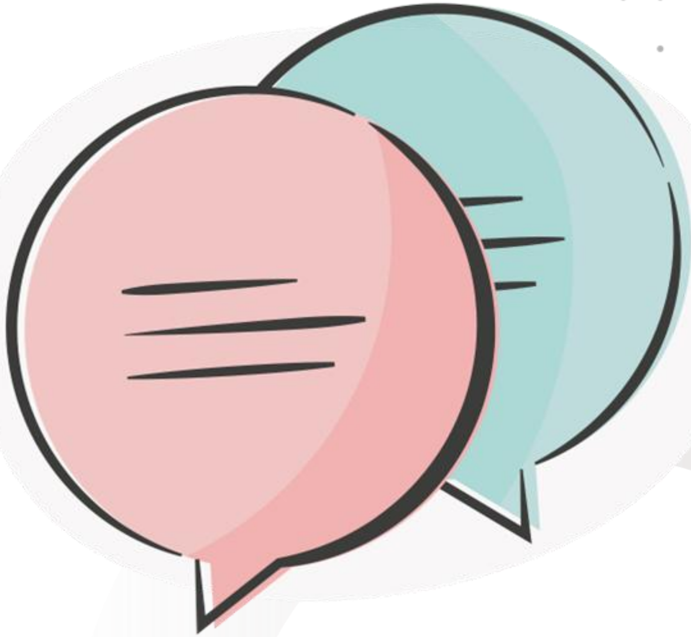
- What are heterogeneous data ingestion patterns?
- Recall key technical considerations when designing ingestion systems.
- What are the main architecture types for heterogeneous data ingestion?
- Examples of tools dealing with heterogeneous ingestion?



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# Intricacies of heterogeneous data ingestion

So far, we have discussed the following aspects of data ingestion:

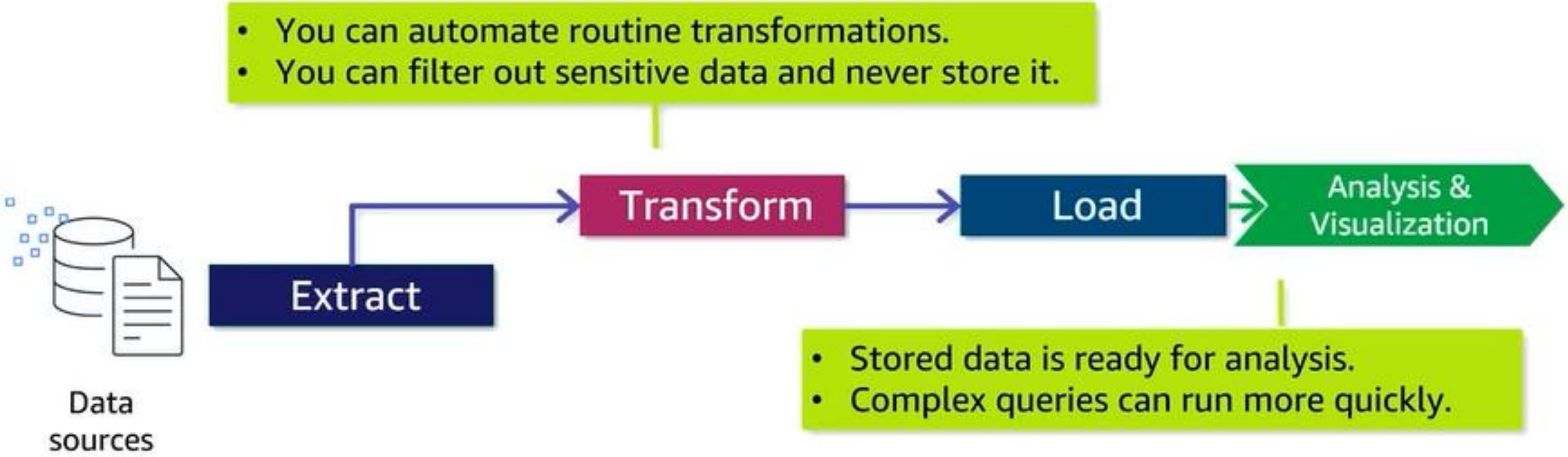
- Data wrangling and cleaning
- Data structuring
- Data validation

However, there is more to heterogeneous data ingestion, and we will now discuss:

- ETL, ELT and Reverse ETL
- Data discovery and cataloguing
- Matching data sources to the right ingestion service

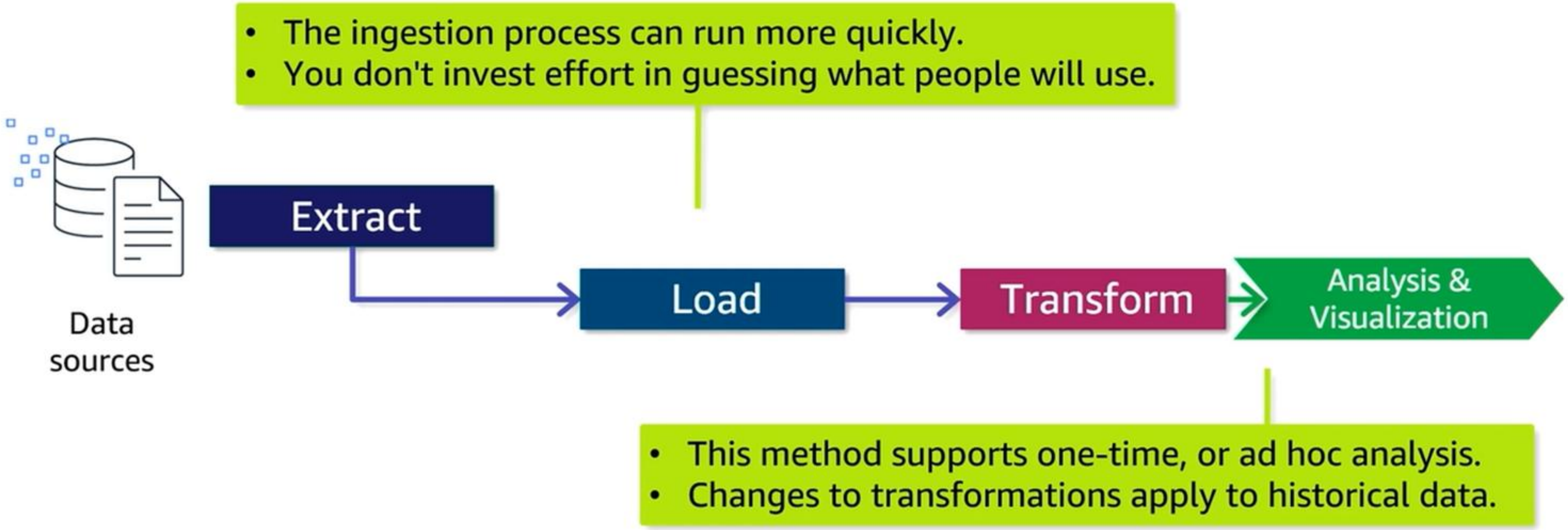
# ETL



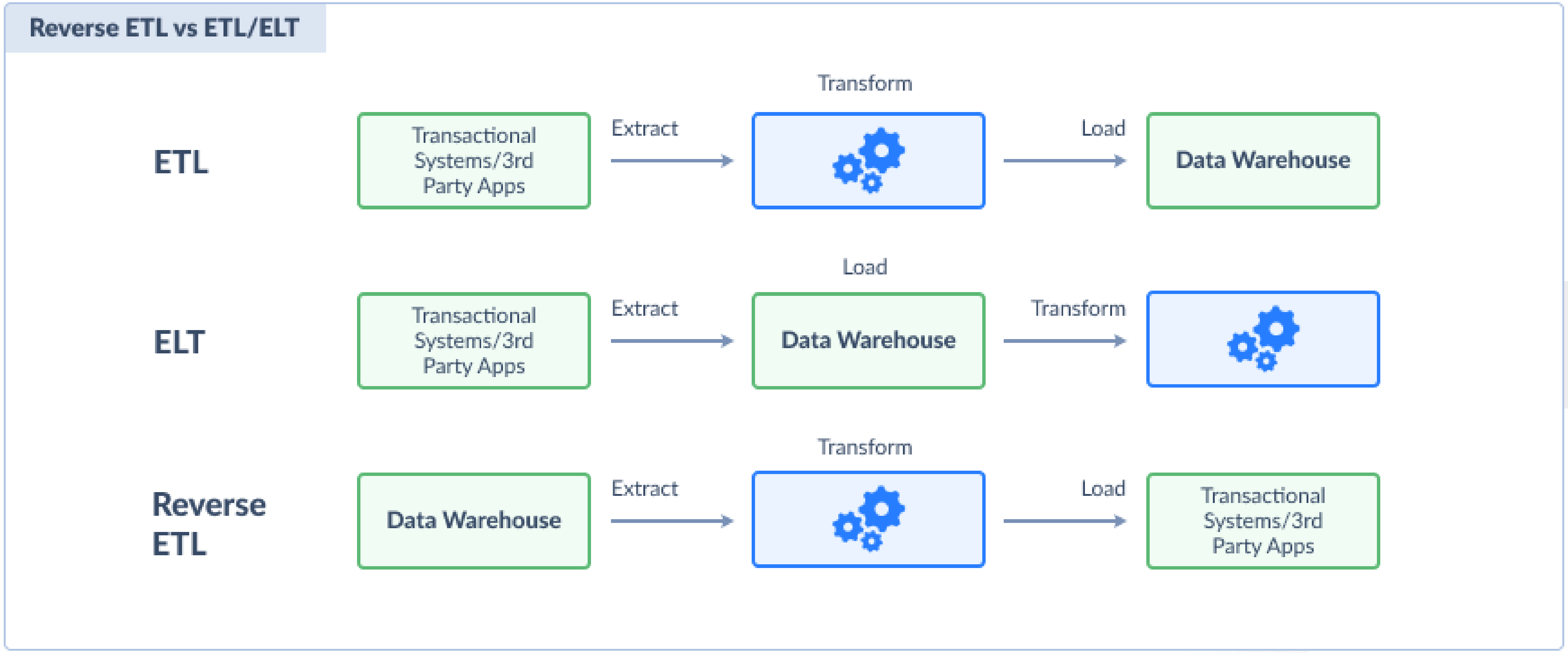

You can automate routine transformations.
You can filter out sensitive data and never store it.
Data sources
Extract
Transform
Load
Analysis & Visualization
Stored data is ready for analysis.
Complex queries can run more quickly.

# ELT




• The ingestion process can run more quickly.
• You don't invest effort in guessing what people will use.
Data sources
Extract
Load
Transform
Analysis & Visualization
• This method supports one-time, or ad hoc analysis.
• Changes to transformations apply to historical data.

# Reverse ETL

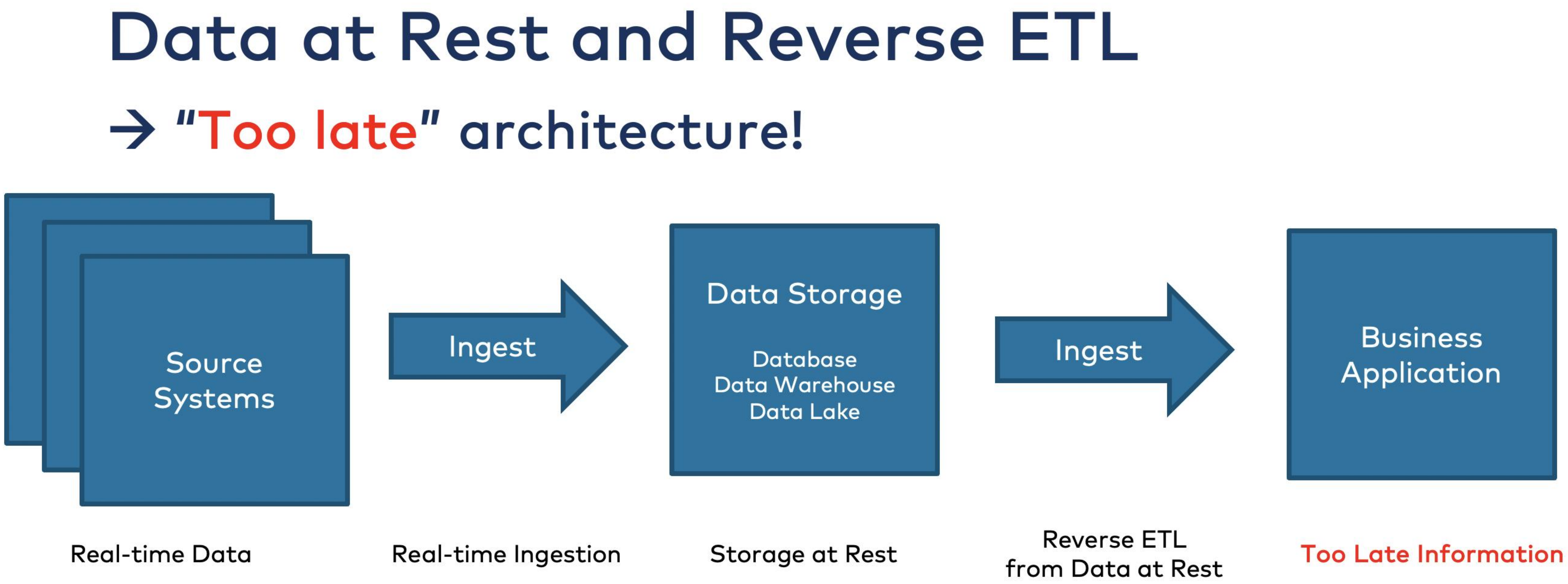


Reverse ETL vs ETL/ELT
ETL
Transactional Systems/3rd Party Apps
Extract
Transform
Load
Data Warehouse
ELT
Transactional Systems/3rd Party Apps
Extract
Load
Data Warehouse
Transform
Reverse ETL
Data Warehouse
Extract
Transform
Load
Transactional Systems/3rd Party Apps

# Use cases for Reverse ETL

- **Identify customers at-risk** and potential customer churn before it happens
- **Drive new sales** by correlating data from the CRM and other interfaces
- **Hyper-personalised marketing** for cross-selling and upselling to existing customers
- **Operational analytics** to monitor the changes in business applications and data faster
- **Data replication to modern cloud applications** for better reporting capabilities and finding insights

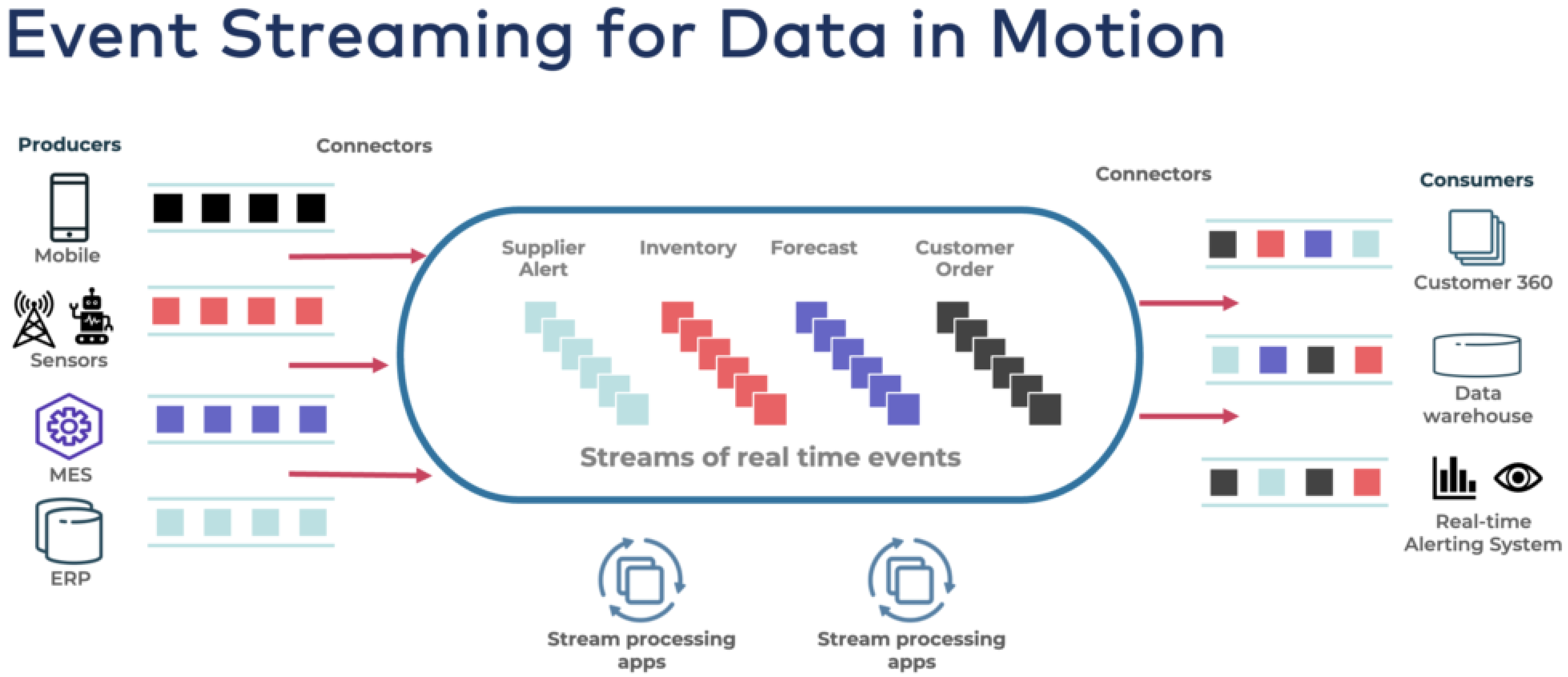
# Reverse ETL can be an anti-pattern




- **Good architecture should never have the goal to plan for reverse ETL from the beginning!**
- It is only needed in brownfield architecture where the data is stored at rest instead of building an event-based architecture

# Reverse ETL

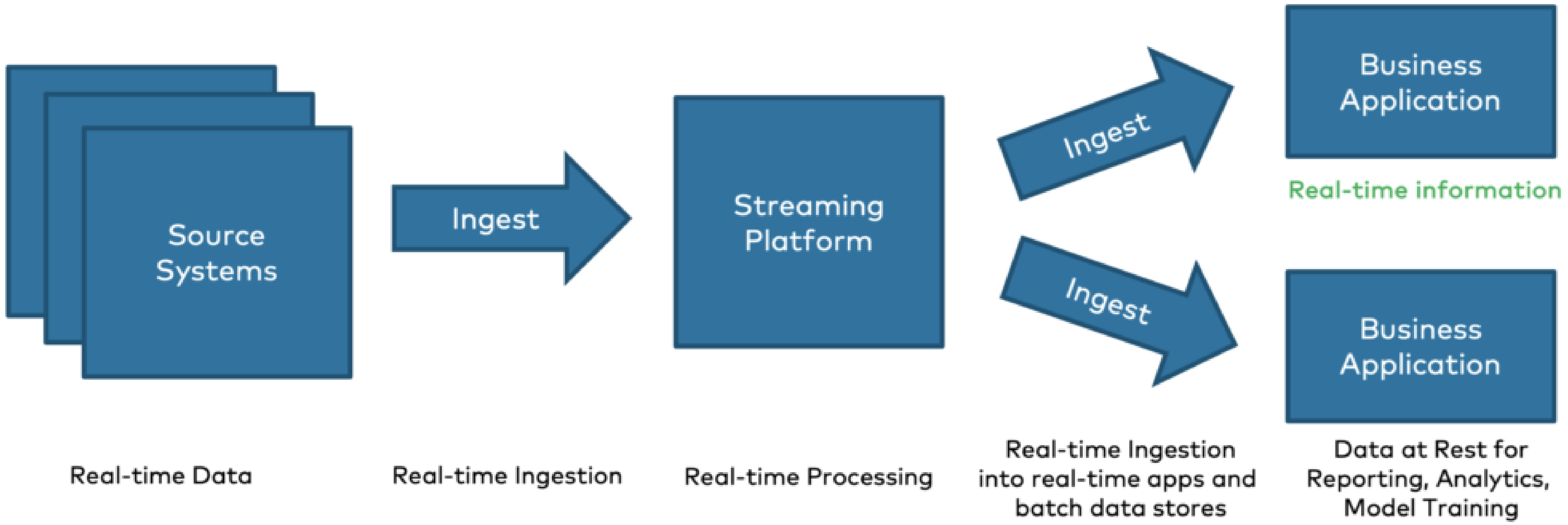
Not needed in modern event-driven architectures…




*Event streaming enables data integration in real-time by nature.*

# Discussion activity

As a Data Engineer, why would you want to maximise transformations on data in motion and minimise data at rest? Is the following an "appropriate" architecture?



# Importance of Data Cataloguing

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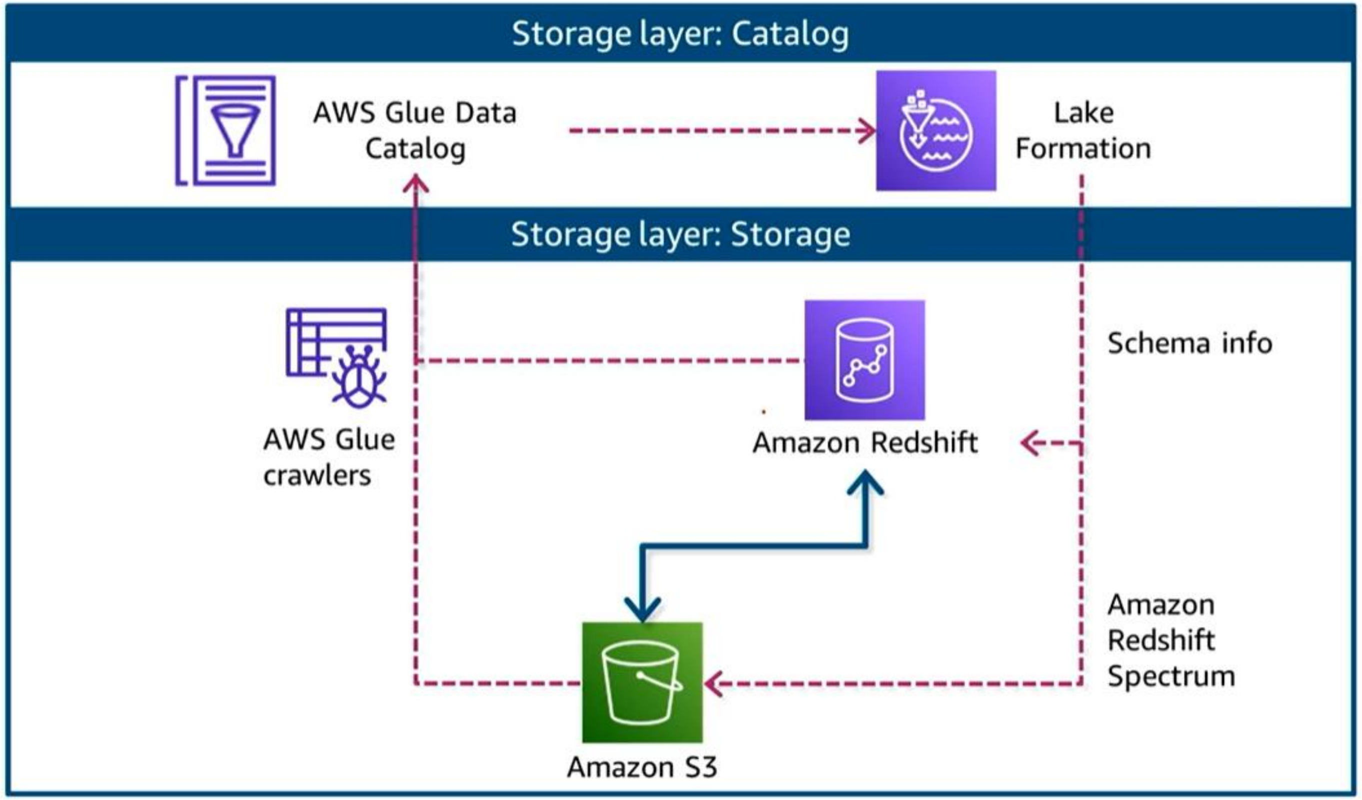


- As organisations collect and process more data, it becomes increasingly important to track and manage metadata effectively.
- Cataloguing provides essential information about data assets, including source, structure and usage.

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# Metadata Catalogues

A metadata catalogue (or a **data catalog**) serves as a repository for knowledge of the data within the mesh. It can help analysts answer important questions about the data such as:

- Where is the database that contains our online order information?
- What is the meaning of this very obscure looking column name?
- What is the quality of this data? Is this data fresh or stale?

- Catalogs help manage regulation compliance and provide tools for tagging sensitive data.
- They offer visibility into which resources are safe to share, and which are not.

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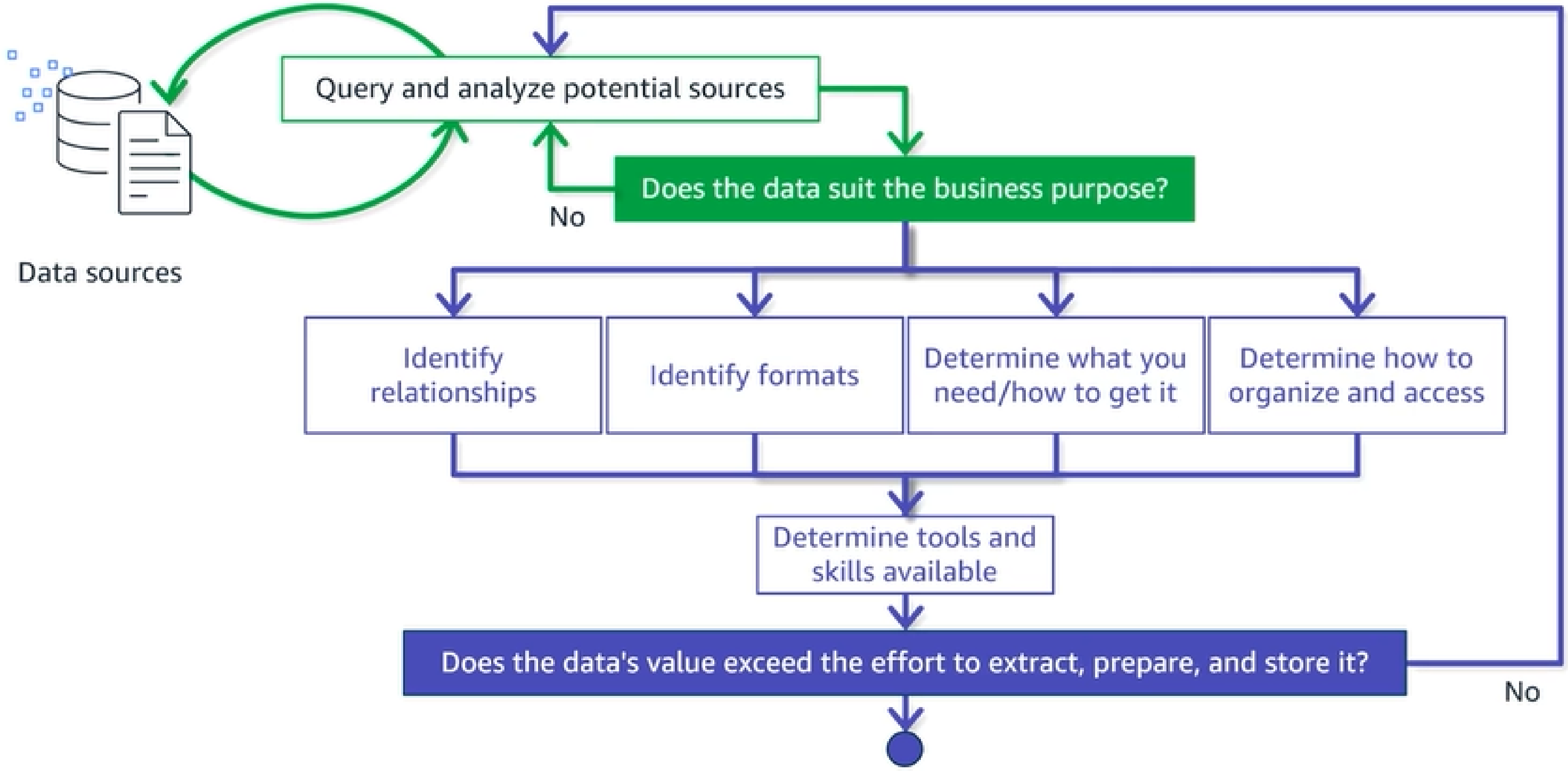
# Cataloguing enhances Discovery


- Efficient data cataloging enhances discovery, which is crucial for organisations to derive value from their data assets.
- Users can easily search, browse, and explore relevant data assets, saving valuable time and effort.
- This speeds up new employee onboarding but also serves as a time saver for existing employees.
- Efficient discovery can also generate new opportunities for different departments to find new and exciting uses for existing data.
- Furthermore, discovery of data lineage enables users to assess the impact of changes, evaluate data quality, and make informed decisions based on reliable and trustworthy data.

# Important considerations at Discovery stage



# Important considerations at Discovery stage

## Scenario analysis

### Identify relationships

- supp1 **requestor_id** = supp2 **cust_num** = customer **customer_id**
- supp1 **ticket_id** = supp2 **issue_id**
- supp1 **subject** = supp2 **description**
- supp1 **priority** "High", "Medium", "Low" = supp2 **priority** 1, 2, 3

### Identify formats

- Ticket systems export .json files
- Storage target = S3 bucket

### Determine what you need/how to get it

- Ticket **status** = "open", "solved", "closed"
- Tickets where **create_date** or **updated_date** = **2020**

### Determine how to organize and access

- Access is by **sales region**
- Create sales region prefixes in an S3 bucket
- Analysts will query with Amazon Redshift Spectrum or Amazon Athena

### Determine tools and skills available

- Based on the exploratory nature of the request, it needs a one-time extract with minimal automation
- The easiest option available is to use Microsoft Excel to wrangle the data
- Excel can import .json and export .csv files

# Architecture of Metadata Catalogues

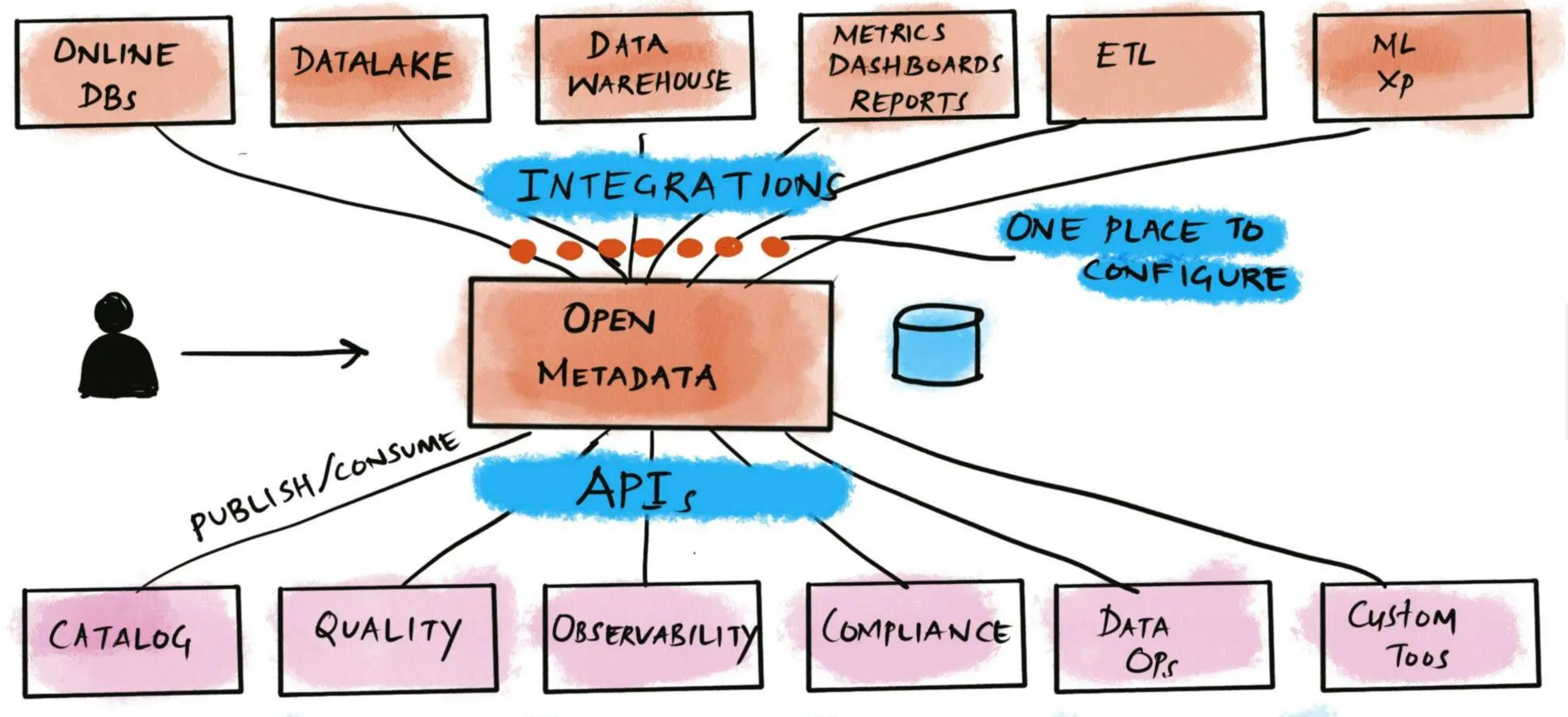
1. **Metadata Store**: This component serves as the central repository for storing metadata, enabling easy access and retrieval. It is designed to handle large volumes of metadata

2. **Metadata Service**: The metadata service handles the storage, retrieval, and manipulation of metadata, ensuring its integrity and accessibility. It provides a set of APIs and interfaces that allow users to interact with the metadata store effectively

3. **Metadata Ingestion**: various ingestion mechanisms to capture metadata from diverse data sources, including connectors, APIs, and automated crawlers. These ingestion mechanisms are designed to handle different types of data sources, such as databases, file systems, cloud storage, and streaming platforms

4. **Metadata Visualisation**: This component provides intuitive interfaces and visualisations, enabling users to explore and interact with metadata effectively.

# OpenMetadata



# OpenMetadata and its future

OpenMetadata is well-positioned to leverage emerging trends, including:

- **Artificial Intelligence and Machine Learning:** OpenMetadata can employ AI and ML techniques to enhance data discovery, automate metadata generation, and improve data quality.
- **Cloud-Native Architecture:** With the shift towards cloud computing, OpenMetadata can optimise its architecture for seamless integration and scalability in the cloud environment.
- **Privacy and Security:** OpenMetadata can incorporate robust privacy and security measures to address the increasing challenges surrounding data protection and compliance.

# Discussion activity

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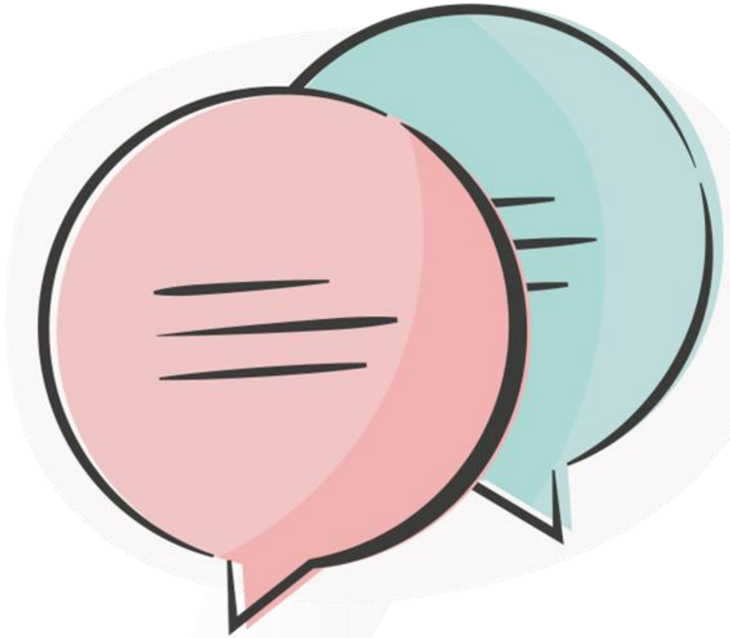
## Reflecting on your own experiences…

- How do you manage metadata at your workplace?
- Share the pros and cons of the approaches that you have seen?



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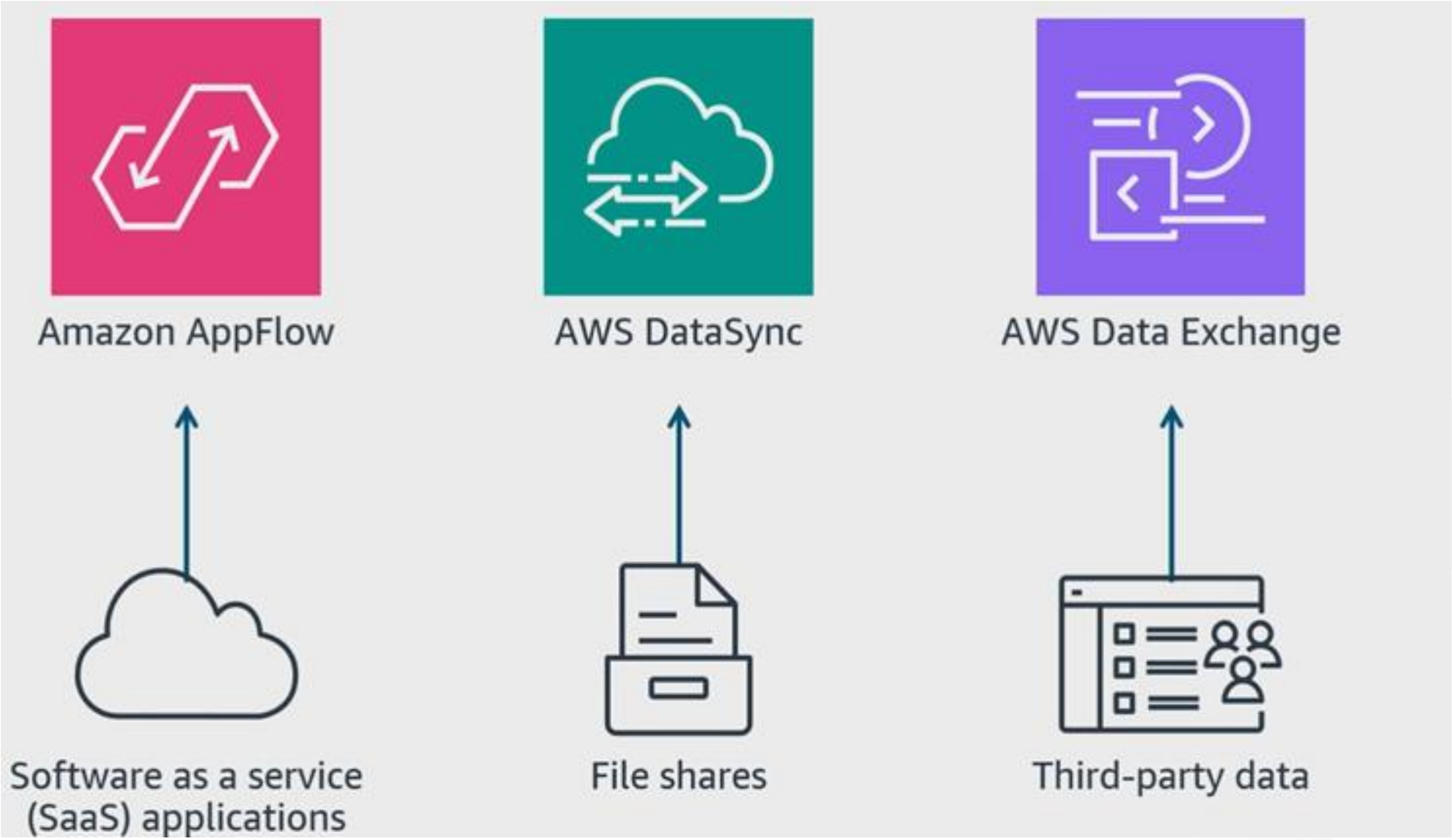
# Matching data sources to ingestion services



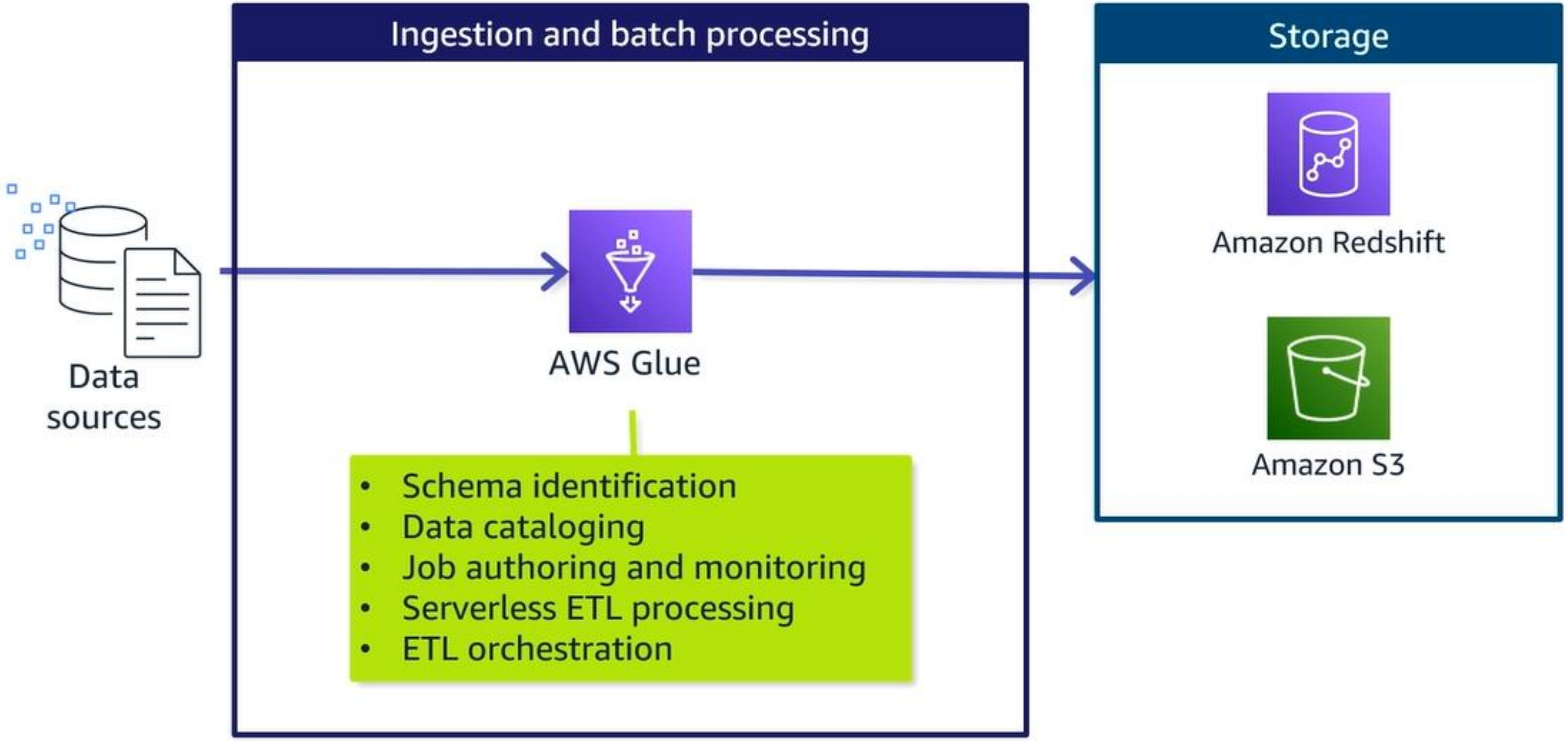
# AppFlow

Use Amazon AppFlow to ingest data from a **software as a service (SaaS)** application.

- Create a connector with filters.
- Map fields and perform transformations.
- Perform validation.
- Securely transfer to Amazon S3 or Amazon Redshift.

Example: Ingest customer support ticket data from Zendesk.

# Using Glue to simplify Batch Ingestion



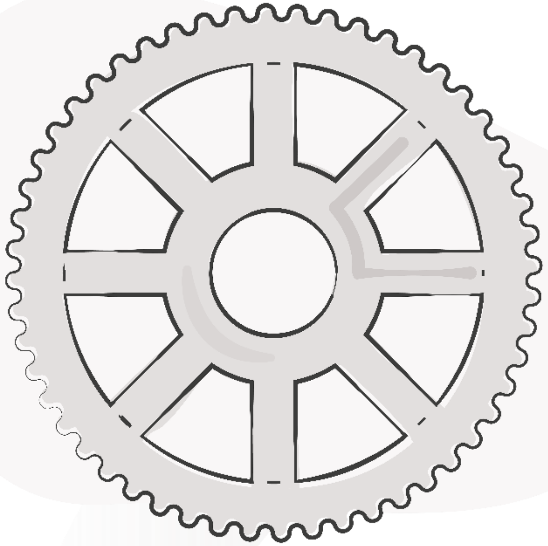
# Practical application

Your tutor will now walk you through **AWS Academy Lab: Performing ETL on a Dataset by using AWS Glue**

*Big data problems often involve a large number of heterogeneous data sources.*

*As a data engineer, you might not know the schema for some data sources. This is the variety aspect of the five Vs of big data (volume, variety, velocity, veracity, and value).*

*In this lab, you will work with AWS Glue to perform extract, transform, and load (ETL) for a dataset. You can direct AWS Glue to a data source, and it can infer a schema based on the data types that it discovers. Then, AWS Glue builds a Data Catalog that contains metadata about the various data sources.*



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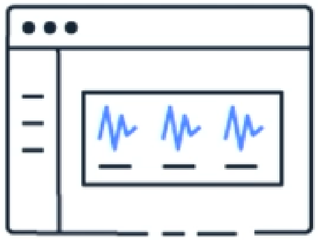
# Things we learned from the lab

- You can build a crawler to discover the schema of a newly ingested dataset and then correctly extract the data from the dataset.
- You can transform the schema of a flat file and then load it into a database.
- You can then analyse the data by using SQL statements.
- You should be mindful of security considerations and always review an Identity and Access Management (IAM) policy for users.

# Things we learned about AWS specifically

- The AWS modern data architecture uses purpose-built tools to ingest data based on characteristics of the data.
- The storage later uses Amazon Redshift as its data warehouse and Amazon S3 for its data lake.
- The Amazon S3 data lake uses prefixes or individual buckets as zones to organise data in different states, from landing to curated.
- AWS Glue and Lake Formation are used in a catalogue later to store metadata.
- With the catalogue, Amazon Redshift Spectrum can query data in Amazon S3 directly.

# Final consideration – data publishing



## Move data into permanent storage

- Apply file management techniques (formats, compression, organization).
- Securely transfer data from source to destination.
- Write the dataset or datasets to the correct storage locations.

## Make data available to consumers

- Apply appropriate access controls to data.
- Save metadata about the dataset.
- Configure how often and when to run the extract.
- Log the update history.

# Putting it all together

## Data sources

- SaaS Applications
- Social Media
- Databases
- CRM and ERM Systems

### Data Ingestion

Data from various source systems arrive at this layer. This layer usually contains connectors for connecting to multiple data sources.

### Data Collection

Data is stored in a temporary staging area like a data mart before processing.

### Data Processing

Cleaning, aggregation, removal of bad data, and deduplication occurs in this layer to produce high-quality data for storage.

### Data Storage

Data is transported to central storage locations like data lakes of warehouses for querying purposes.

### Data Querying

SQL, ML and BI tools perform data operations on stored data.

### Data Visualization

Insights are presented to users/ business decision-makers in the form of graphs, dashboards, tables, maps, etc.

## End data use-cases

- Analytics
- Machine Learning
- Business Intelligence & Reporting
- Data Science

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# Key Learning Summary

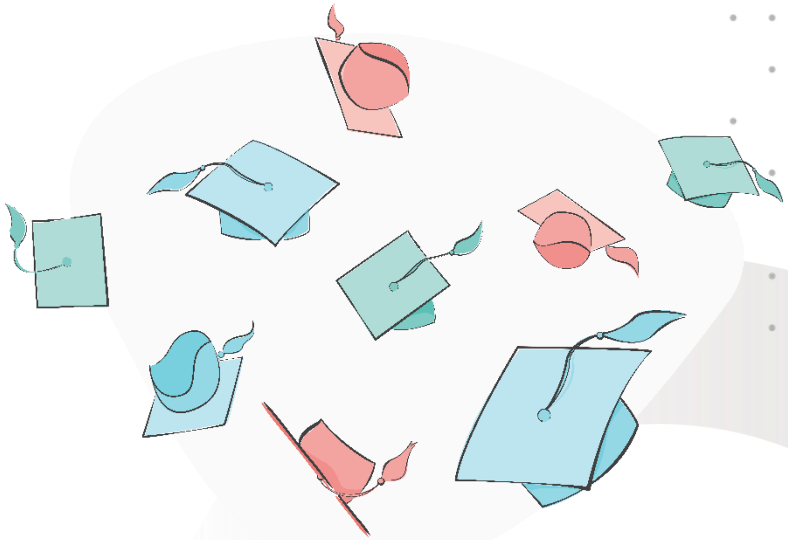
**The key takeaways from this session are as follows:**

- **Automation in Data Collection and Ingestion:** Automation reduces errors and increases efficiency in data handling.

- **Data Cleaning Techniques:** Common techniques include removing duplicates, handling missing values, and standardising data formats.

- **Pre-Processing for Machine Learning:** Pre-processing steps include normalisation, encoding categorical variables, and feature scaling.

- **Practical Skills:** Hands-on experience with tools like Python and pandas for data manipulation and analysis is crucial.



# Thank you

## Do you have any questions, comments, or feedback?